

Dairi Oussama Abderrahmane

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EDUCATION

École supérieure en informatique de Sidi Bel Abbès

12/2020 – 06/2022

École Nationale Supérieure d'Informatique

09/2022 – 09/2025

Engineer's degree, Master's degree, Computer Science, Option: Computer Systems (SIQ)

Top Courses : Advanced Networking, Cybersecurity, High-performance Computing, Distributed Systems

PROFESSIONAL EXPERIENCE

Research Intern

09/2024 – 07/2025

Laboratoire Informatique, Image et Interaction (L3i)

La Rochelle, France

- Proposed and evaluated a two-agent framework for autonomous network-layer exploitation named **INVEST**.
- Leveraged **DeepSeek-V3** with role-constrained prompt engineering to enable protocol-aware planning and bounded adaptive replanning.
- Evaluated system across **6 packet-level attack scenarios (L2–L7) and 1 CVE**.
- Engineered packet-level KPI validation using isolated VM testbeds (Kali Linux, Ubuntu, Linux Mint) with reproducible benchmarks.

Back-end Developer

10/2025 – Present

Remote

- Designed and developed RESTful backend API using Django REST Framework and MySQL.
- Implemented role-based authorization mechanisms to enforce secure resource access.
- Conducted manual security audits identifying and mitigating potential DoS vectors and authorization flaws.

Web development intern

07/2023 – 09/2023

IRIS / EURL SATEREX

Setif, Algeria

Third year Internship at IRIS (Algerian electronics firm) as a web developer. Developed a web app to manage employee leave requests and approval workflows with role-based access control.

RELATED EXPERIENCE

CTF Player

2023 – present

CTF (Capture The Flag) competitions consist in competitive challenges that cover various topics of cybersecurity (web security, binary exploitation, cryptography, digital forensics, etc.)

- Competed in multiple national and international CTF competitions specializing in binary exploitation and reverse engineering.
- Ranked Top 1% globally on Root-Me (Rank: 1178).
- Authored technical write-ups detailing different types of security exploitation that can be found [here](#).
- I participate with both teams **noreply** and **th3_ronins**.

Cyber Security team leader

10/2023 – 06/2024

GDG Algiers

GDG Algiers is a local community and student's club situated at the Higher National School of Computer Science in Algiers (ESI ex INI), Algiers, Algeria. It is part of the big global community of developers "Google Developers Group".

- I was responsible for the club's cyber security related activities.

PROJECTS

INVEST

- A two-agent LLM framework (Supervisor–Executor) to autonomously plan and execute packet-level attacks across OSI Layers 2–7.

- Implemented bounded feedback-loop control using stdout/stderr parsing and packet-level KPI validation for adaptive replanning.
- Designed reproducible benchmark of 7 network attack scenarios (flooding, ARP spoofing, TCP hijacking, CVE exploitation) in isolated VM testbeds.
- Paper under submission (2026)

Development of security challenges for CTF competitions

Authored binary exploitation and reverse engineering challenges for national CTF competitions (HackINI 2k24, DevFest 2023).

Network Configuration and SIEM Integration

For the fourth year project, I was in a team of 6 students that :

- Designed and simulated a network infrastructure using GNS3 and VMware.
- Configured pfSense firewall, Deployed Snort IDS and integrated Wazuh SIEM into the infrastructure.
- Simulated network attacks to validate intrusion detection capabilities.
- Implemented log aggregation and rule-based alerting mechanisms.

Back-End Projects

I worked on many web development projects, specifically on the back-end, some of these projects include a train ticket reservation system, a social media for people suffering from addiction, a platform for renting/buying houses and a platform to help tourists navigate better.

SKILLS

Programming and scripting

Python, Bash, C, JavaScript, PHP, Java

Cybersecurity

Web security, Binary Exploitation, Reverse Engineering, SIEM, Network Security

Frameworks & Libraries

Django/DRF, Flask, FastAPI, Laravel

Tools & Platforms

Ghidra, GDB, Wireshark, Burp Suite, GNS3, VirtualBox, Linux, Git/GitHub

LLMs & Agents

Multi-agent orchestration, RAG, prompt engineering

LANGUAGES

Arabic	● ● ● ● ●	French	● ● ● ● ●
English	● ● ● ● ●	C1	